Scoring Physician Risk Communication in Prostate Cancer Using Large Language Models

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Introduction

Effective risk communication is essential in prostate cancer care, where treatment decisions involve complex tradeoffs between similar oncologic outcomes and differing side effect profiles. Shared decision-making (SDM) relies on physicians clearly conveying key risks and prognostic information, yet real-world communication quality remains inconsistent. Manual evaluation of risk communication is labor-intensive and not scalable.

We propose an automated framework that uses large language models (LLMs) and rubric-based prompting to score the quality of physician communication about five key concepts central to SDM: Cancer Prognosis (CP), Life Expectancy (LE), Erectile Dysfunction (ED), Urinary Incontinence (INC), and Irritative Urinary Symptoms (IUS).

Our method moves beyond identifying whether these concepts were mentioned, assessing instead the specificity, clarity, and patient-centeredness of the communicated information using fine-grained quality scores (0–5).

Methods

We analyzed transcripts from 20 real-world prostate cancer consultations. For each of the five concepts (CP, LE, ED, INC, IUS), we used a validated NLP model to extract high-yield text segments (7-sentence fragments) likely to contain relevant risk communication content.

Expert annotators assigned quality scores from 0 to 5 using validated rubrics for each concept. Higher scores reflect greater quantitative precision and personalization.

We modeled each scoring task as a 6-class text classification problem. Two prompting strategies were tested with GPT-4o:

- **Rubric-only prompting**
- **Rubric + Chain-of-Thought (CoT) prompting for structured reasoning**

We compared LLM-based approaches to fine-tuned BERT-based classifiers (RoBERTa-large and PubMedBERT-large), reporting micro-averaged F_1 scores for each concept.

Table 1. Scoring rubric for Cancer Prognosis, Life Expectancy, and Side Effects. The rubrics of *Side Effects* column apply to ED, INC, and IUS.

Score	Cancer Prognosis	Life Expectancy	Side Effects
0	Not mentioned	Not mentioned	Not mentioned
1	Mention without quantification (e.g., general risk mentioned)	Mention without estimation	Name only (e.g., "erectile dysfunction")
2	Qualitative estimate only (e.g., "low risk")	Qualitative estimate only (e.g., "long life expectancy")	Qualitative terms only (e.g., "unlikely to affect you")
3	Numeric estimate without treatment comparison	Rough number of years (e.g., "20–30 years")	Numeric estimate without timeline (e.g., "30% will experience it")
4	Numeric estimate including risk with and without treatment	Probability of survival/mortality at a timepoint	Numeric estimate with timeline (e.g., "10% at 1 year")
5	Patient-specific numeric estimate at life expectancy, with and without treatment	Specific estimate based on patient factors (e.g., age, health status)	Patient-specific numeric estimate with timeline and reference to individual characteristics

Results

We evaluated all models using micro-averaged F_1 scores across the five SDM concepts. The best-performing method, GPT-4o with rubric + CoT prompting, achieved micro-averaged F_1 scores between 85.0 and 92.0 across all domains, outperforming traditional models and matching expert agreement. Adding few-shot examples to the prompts consistently improved performance over zero-shot variants, and incorporating CoT reasoning further enhanced accuracy, particularly for nuanced, multi-step risk assessments.

Table 5. Micro-averaged F_1 scores (%) for baseline classifiers and LLM-based prompting strategies across the five key concepts. The best-performing method for each concept is shown in bold.

Approach	Model/Setting	CP	LE	ED	INC	IUS
Baseline	RoBERTa-large	42.00	73.33	41.67	53.33	78.33
	PubMedBERT-large	52.00	63.33	45.00	41.67	68.33
LLM (Rubric)	Zero-shot	58.00	71.67	60.00	63.33	78.33
	Few-shot	76.00	86.67	73.33	65.00	85.00
LLM (Rubric + CoT)	Zero-shot	78.00	76.67	70.00	78.33	88.33
	Few-shot	**92.00**	**86.67**	**85.00**	**85.00**	**91.67**

Discussion

Our findings demonstrate that LLMs, particularly when guided with domain-specific rubrics and CoT reasoning, can accurately and interpretably evaluate physician–patient risk communication, even with limited annotated data.

- **CoT prompting** improved transparency by explicitly breaking down the reasoning behind each score, an important feature for trust and clinical integration.
- The LLM's performance **matched human annotator agreement** ($F_1 \approx 85$–87), suggesting it can function at near-expert level.
- Most model errors involved **adjacent score confusions** (e.g., 2 vs 3), often tied to vague or implicit language, issues that also challenged expert human annotators.

Future work will expand to larger, multi-institutional datasets and develop encounter-level scoring methods for deployment in real-time clinical feedback systems.

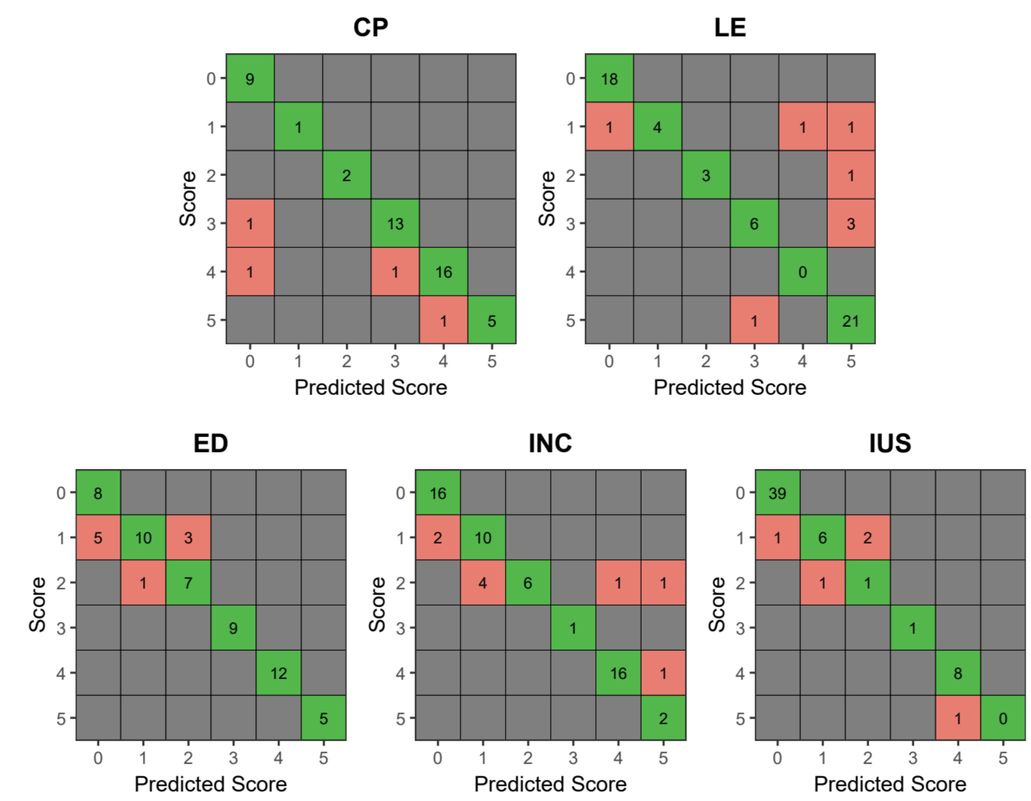

Fig. 1. Confusion matrices for each of the five key concepts using the best-performing system (rubric + CoT with few-shot prompting). Each matrix displays the distribution of predicted versus true scores (0–5).

Conclusion

We present a novel framework for evaluating the quality of physician risk communication in prostate cancer consultations using LLMs. By combining validated scoring rubrics with CoT prompting, our method enables interpretable, sentence-level assessment of how precisely and patient-specifically key concepts are communicated. Across five clinically relevant domains, including cancer prognosis, life expectancy, and three side effects, our best-performing model achieved strong results and expert-level agreement. These findings highlight the potential of LLMs to support scalable, automated evaluations of physician–patient communication and lay the groundwork for real-time feedback tools to improve shared decision-making in oncology and beyond.

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